Certified SF libraries for zero-shot planning in stochastic domains

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Question. Given a set of successor features, is GPI over them ϵ-optimal for every weight, so that any new FSA task can be planned zero-shot? If not, which policy must be added, and how can it be obtained without training it? We answer both with Bellman gaps only: no linear system is solved, every quantity is built from the one-step SF backup, which has a sample-based (TD) counterpart.

1 Setting

- MDP: finite $\mathcal{S}, \mathcal{A}$; $P(s' \mid s, a) \geq 0$, $\sum_{s'} P(s' \mid s, a) \leq 1$ (missing mass: episode ends); $\boldsymbol{\phi}(s, a) \in \mathbb{R}^d_{\geq 0}$ expected features; $\gamma \in (0, 1)$. Weights $\mathbf{w} \in \mathbb{R}^d_{\geq 0}$, $\Delta = \{\mathbf{w} \geq 0 : \sum_i w_i = 1\}$.
- Policy $\pi : \mathcal{S} \to \mathcal{A}$. SF backup $(T^\pi \boldsymbol{\psi})(s, a) = \boldsymbol{\phi}(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, \boldsymbol{\psi}(s', \pi(s'))$; the action SFs $\boldsymbol{\psi}^\pi$ are its fixed point. $Q^\pi_{\mathbf{w}}(s, a) = \boldsymbol{\psi}^\pi(s, a)^\top \mathbf{w}$, $V^\pi_{\mathbf{w}}(s) = Q^\pi_{\mathbf{w}}(s, \pi(s))$, $V^*_{\mathbf{w}} = \max_\pi V^\pi_{\mathbf{w}}$.
- Library L: a finite set of policies with their action SFs. GPI estimate and GPI policy [2]:

$$\hat{Q}^L_{\mathbf{w}}(s, a) = \max_{\pi \in L} \boldsymbol{\psi}^\pi(s, a)^\top \mathbf{w}, \qquad \pi^G_{\mathbf{w}}(s) \in \arg\max_a \hat{Q}^L_{\mathbf{w}}(s, a), \qquad V^{\pi^G_{\mathbf{w}}}_{\mathbf{w}} \geq \max_a \hat{Q}^L_{\mathbf{w}}(\cdot, a).$$

- Relevant states $\mathcal{R}$: the start state and the exits (SF-FSA-VI reads values and starts subtasks only there).
- Weight region: a union of faces $\Omega = \bigcup_{F \in \mathcal{F}} \Omega_F$, $\Omega_F = \{\mathbf{w} \in \Delta : \mathrm{supp}(\mathbf{w}) \subseteq F\}$. A class of FSA tasks fixes $\mathcal{F}$: an automaton state that accepts the symbols Σ' puts weight only on their exits, so states branching on $\leq m$ symbols need the faces of m symbols; $\mathcal{F} = \{\{1..d\}\}$ covers every FSA.
- SF-FSA-VI [1], zero-shot: each automaton state u gets $\mathbf{w}(u) \in [0, 1]^d$ with $w_e(u) = 1$ if exit e ends the task, $w_e(u) = \max_a \hat{Q}^L_{\mathbf{w}(v)}(e, a)$ if it leads to v, 0 otherwise (reverse topological order); the agent executes $\pi^G_{\mathbf{w}(u)}$ in u.

2 Solve-free primitives

Evaluation. $\boldsymbol{\psi}_{k+1} = T^\pi \boldsymbol{\psi}_k$ from any start $\boldsymbol{\psi}_0$; $\|\boldsymbol{\psi}_k - \boldsymbol{\psi}^\pi\|_\infty \leq \frac{\gamma}{1-\gamma} \|\boldsymbol{\psi}_k - \boldsymbol{\psi}_{k-1}\|_\infty$. With samples: TD on stored transitions $(s, a, \boldsymbol{\phi}, s')$. Cost unit: one *sweep* = one backup of all (s, a).

Bellman gap. $g^\pi_{\mathbf{w}}(s) = \max_a Q^\pi_{\mathbf{w}}(s, a) - V^\pi_{\mathbf{w}}(s) \geq 0$; read off the action-SF table.

Reachable layers. $\mathcal{L}_0(r) = \{r\}$, $\mathcal{L}_{t+1}(r) = \{s' : \exists s \in \mathcal{L}_t(r), a,\ P(s' \mid s, a) > 0\}$; they become constant after some t_0.

Proposition 1 (gap bound)**.** *For any π, $\mathbf{w}$, r:*

$$g^\pi_{\mathbf{w}}(r) \ \le\ V^*_{\mathbf{w}}(r) - V^\pi_{\mathbf{w}}(r) \ \le\ B^\pi_{\mathbf{w}}(r) := \textstyle\sum_{t\ge 0} \gamma^t \max_{s\in\mathcal{L}_t(r)} g^\pi_{\mathbf{w}}(s).$$

More generally, for any $V : \mathcal{S} \to \mathbb{R}$ with residual $\rho(s) = \max_a[r_{\mathbf{w}}(s,a) + \gamma \sum_{s'} P(s' \mid s, a)V(s')] - V(s)$: $V^*_{\mathbf{w}}(r) - V(r) \le \sum_t \gamma^t \max_{\mathcal{L}_t(r)} \rho^+$.

Proof. Upper: $V^* - V = (T^{\pi^*}V^* - T^{\pi^*}V) + (T^{\pi^*}V - V) \le \gamma P_{\pi^*}(V^* - V) + \rho$; iterate; row r of $P^t_{\pi^*}$ is a sub-probability on $\mathcal{L}_t(r)$. With $V = V^\pi$, $\rho = g^\pi$. Lower: $V^*(r) \ge \max_a[r_{\mathbf{w}}(r,a) + \gamma \sum PV^\pi] = V^\pi(r) + g^\pi(r)$. □

3 The certificate

Definition 1 (records)**.** A record $(\mathbf{w}_q, \mathbf{u}_q)$, $\mathbf{w}_q \in \Omega_F$, $\mathbf{u}_q \in \mathbb{R}^{\mathcal{R}}$, satisfies $u_q(r) \ge V^*_{\mathbf{w}_q}(r)$ for all r. Any evaluated policy π gives one: $u_q(r) = V^\pi_{\mathbf{w}_q}(r) + B^\pi_{\mathbf{w}_q}(r)$ (Proposition 1).

For a face F, $r \in \mathcal{R}$ and $\mathbf{w} \in \Omega_F$ define (records restricted to Ω_F)

$$E_r(\mathbf{w}) = \max_{\pi\in L,\, a\in\mathcal{A}} \boldsymbol{\psi}^\pi(r,a)^\top \mathbf{w},$$

$$U^F_r(\mathbf{w}) = \min\Big\{ \textstyle\sum_q \lambda_q u_q(r) :\ \sum_q \lambda_q \mathbf{w}_q = \mathbf{w},\ \lambda \ge 0,\ \mathbf{w}_q \in \Omega_F \Big\}.$$

E_r is convex and piecewise affine; its *corners* C^F_r are the vertices of its linearity regions in Ω_F (the vertices of $\{\mathbf{w} \in \Omega_F : \boldsymbol{\psi}^\pi(r,a)^\top \mathbf{w} \ge E_r(\mathbf{w})\}$ over all (π, a)). U^F_r is the lower convex envelope of the records in Ω_F.

Theorem 1 (certificate)**.** *Let $\Gamma = \max_{F\in\mathcal{F}} \max_{r\in\mathcal{R}} \max_{\mathbf{c}\in C^F_r} [U^F_r(\mathbf{c}) - E_r(\mathbf{c})]$ (corners and records of each face). Then for every $\mathbf{w} \in \mathbb{R}^d_{\ge 0}$ with $\mathbf{w}/\|\mathbf{w}\|_1 \in \Omega$ and every $r \in \mathcal{R}$,*

$$V^*_{\mathbf{w}}(r) - V^{\pi^G_{\mathbf{w}}}_{\mathbf{w}}(r) \ \le\ V^*_{\mathbf{w}}(r) - \max_a \hat{Q}^L_{\mathbf{w}}(r,a) \ \le\ \Gamma\, \|\mathbf{w}\|_1.$$

Proof. First inequality: GPI [2]. $\max_a \hat{Q}^L_{\mathbf{w}}(r,a) = E_r(\mathbf{w})$. $V^*_{\mathbf{w}}(r)$ is convex in $\mathbf{w}$ (maximum of linear functions), so $V^*_{\mathbf{w}}(r) \le U_r(\mathbf{w})$. On each linearity region of E_r, $U_r - E_r$ is convex, so its maximum over the region is attained at a vertex; hence $\max_{\Omega_F}(U_r - E_r) = \max_{C_r}(U_r - E_r) \le \Gamma$. Both sides are positively homogeneous of degree 1 in $\mathbf{w}$. □

The bound covers both what GPI achieves and what the planner believes (the estimate), at every relevant state, for every weight of the face. It needs only the stored SFs at $\mathcal{R}$ and the records.

Corollary 1 (zero-shot FSA)**.** *Assume one-hot exit features (so $\sum_i \psi^\pi_i(s) \le 1$). For an acyclic task of depth H (longest path from u_0) whose weights $\mathbf{w}(u)$ lie in the certified face, with $m = \max_u \|\mathbf{w}(u)\|_1$, the zero-shot policy loses at most $(H+1)^2\, m\, \Gamma$ with respect to the optimal policy of SF-FSA-VI.*

Sketch. Let $\delta_u = \|\mathbf{w}(u) - \mathbf{w}^*(u)\|_\infty$ and ε_u the worst loss from u at relevant states. Lipschitzness of $V^*_{\mathbf{w}}$ in $\mathbf{w}$ (constant 1) and Theorem 1 give $\delta_u \le \max_v(\delta_v + m\Gamma)$ and $\varepsilon_u \le \delta_u + m\Gamma + \max_v(\delta_v + m\Gamma + \varepsilon_v)$ over successors v; induction on depth. □

4 Building a certified library

L is split into the *base* B (policies obtained by control) and *compositions* (obtained by evaluating a policy built from L). A corner $(\mathbf{c}, r)$ is *closed* when $U_r(\mathbf{c}) - E_r(\mathbf{c}) \leq \theta$ ($\theta = 10^{-9}$: exact mode).

Algorithm 1 Certified library (CL)

Require: face F, tolerance ϵ

$B = L \leftarrow \{\pi^*_{\mathbf{e}_i} : i \in F\}$ (control); records at the vertices
loop
  $(\mathbf{c}, r) \leftarrow$ corner with the largest gap (lazy max-heap; gaps only decrease)
  if gap $\leq \epsilon$ **then return** L ▷ $\Gamma \leq \epsilon$
  end if
  FREE: record $\mathbf{u} = \min\left(\min_{\pi \in L}[V^\pi_{\mathbf{c}} + B^\pi_{\mathbf{c}}],\ \hat{V} + \sum_t \gamma^t \max_{\mathcal{L}_t} \rho^+\right)$ with $\hat{V} = \max_a \hat{Q}^L_{\mathbf{c}}$ ▷ stored SFs; 1 backup
  if closed: **continue**
  COMPOSE: evaluate $\pi^G_{\mathbf{c}}$ (warm start $\hat{Q}^L$); **if** $B^{\pi^G}_{\mathbf{c}}(r) \leq \theta$: add to L, record, **continue**
  META: policy iteration restricted to $\{\pi(s) : \pi \in B\}$; **if** certified: add to L, record, **continue**
  LEARN: policy iteration over $\mathcal{A}$ from the best candidate (anytime: stop when certified); add to B and L, record
end loop

Proposition 2. *(i) After a corner is answered, its gap is at most θ. (ii) Gaps never increase, so the largest gap is an anytime certificate Γ_t of the current library. (iii) A policy is trained only at a weight where neither the stored SFs, nor GPI's policy, nor the metapolicy over B is certified.*

Sufficiency test. Algorithm 1 without LEARN: a corner that no rung closes is kept open, and the best composition is still added (it is a valid policy with known SFs). It returns the certificate of everything that can be composed from a given base B, and the weights where B is not certified.

Corners. For $|F| = 2$ the corners are breakpoints of an upper envelope of lines (exact). For $|F| \geq 3$ every point's linearity region $\{\mathbf{w} \in \Omega_F : (\boldsymbol{\psi} - \boldsymbol{\psi}')^\top \mathbf{w} \geq 0\ \forall \boldsymbol{\psi}'\}$ is enumerated (Chebyshev centre by LP, vertices by halfspace intersection); points whose region has no interior are discarded. U^F_r is evaluated through the lower facets of the convex hull of the records (or by its LP).

Sample-based version. Replace P by the empirical model $\hat{P}$ of a replay buffer: TD on the buffer converges to the same SFs. Theorem 1 then holds for $\hat{P}$; for P an additional model-error term is needed (Section 6).

5 What a certificate costs

Proposition 3 (informal)**.** *A certificate of the form of Theorem 1 needs, for every $\mathbf{w} \in \Omega$, an evaluated policy in L whose value is within Γ of $V^*_{\mathbf{w}}$ at every r: L must be a Γ-cover of Ω, i.e. at least as large as a Γ-CCS. Compositions replace* training *(control) by* evaluation *(prediction), not the number of evaluated policies. Iterating compositions at one weight is policy iteration (GPI over $L \cup \{\pi^G\}$ is at least the greedy step), so a base of the d extreme policies always suffices in principle; the cost moves into evaluations.*

In deterministic domains with exit features the GPI estimate is already exact ($E_r = V^*$), so every corner is closed by FREE and no composition is needed.

6 Experiments

All runs are exact in P (no sampling) unless stated; costs are SF sweeps; "trained" counts control problems (policy iteration over all actions), "library" all stored SFs; true losses are measured on held-out weights inside the certified region (ground truth by policy iteration). Every run is one anytime run; a row is its state when $\Gamma \leq \epsilon$ is first certified.

Deterministic Office (sanity check). With the 6 extreme policies every corner of the full 6-exit simplex is closed by FREE: $\Gamma = 5.7 \cdot 10^{-8}$ at all 7 relevant states after 383 corner checks, 684 sweeps, no composition and no training. GPI over the extremes is certified optimal for every FSA, recovering the known deterministic result.

Double slit (Kuric et al.'s DoubleSlit, wind ± 3, $\gamma = 0.9$; Table 1, Figure 1). All methods reach the same certificates with libraries of about the same size; the certified library trains 11 policies instead of 309, at the compute of warm-started SFOLS plus 9–20%, and 37× less than SFOLS learning every policy from scratch. The sufficiency test (Table 2) shows that compositions alone, from the 2 extreme policies, already reach an exact certificate. With an empirical model (Table 3) the certificate is only valid for $\hat{P}$: the true loss exceeds it until $n \approx 10^4$ samples per state-action.

Windy two-stage and Office (Tables 4–5, Figures 2, 3). Every certified library plans every task of its class zero-shot with regret below 10^{-5} (windy $6.7 \cdot 10^{-6}$, Office $p = 0.1$ $2.7 \cdot 10^{-6}$). The 6 extreme policies lose up to 1.2% in windy, where the first stage must hedge between the two slits; in Office with $p = 0.1$ they are already near-optimal on tasks ($1.6 \cdot 10^{-5}$) but not on the whole class (worst loss $6.6 \cdot 10^{-4}$ against $9.9 \cdot 10^{-5}$ for the certified library). In windy the FREE rung closes many corners, so the certified library is both smaller and cheaper than SFOLS (15 policies and 2,865 sweeps against 23 and 4,032 at $\epsilon = 10^{-3}$). In Office FREE almost never closes a corner: the three methods certify with libraries of the same size, only 24 of 255 policies are trained, and the certified library costs 5% more sweeps than warm-started SFOLS (failed composition attempts).

Office, $p = 0.3$ (provisional, runs in progress). Same task class, $\epsilon = 2 \cdot 10^{-3}$. At the last checkpoint all three runs certify $\Gamma = 9.2 \cdot 10^{-3}$: certified library 176 policies (21 trained, 36,101 sweeps), Certify-or-Grow 178 (22, 35,326), warm-started SFOLS 178 (178, 33,697). The pattern of $p = 0.1$ repeats: about 8× fewer trained policies, the same library and 7% more compute.

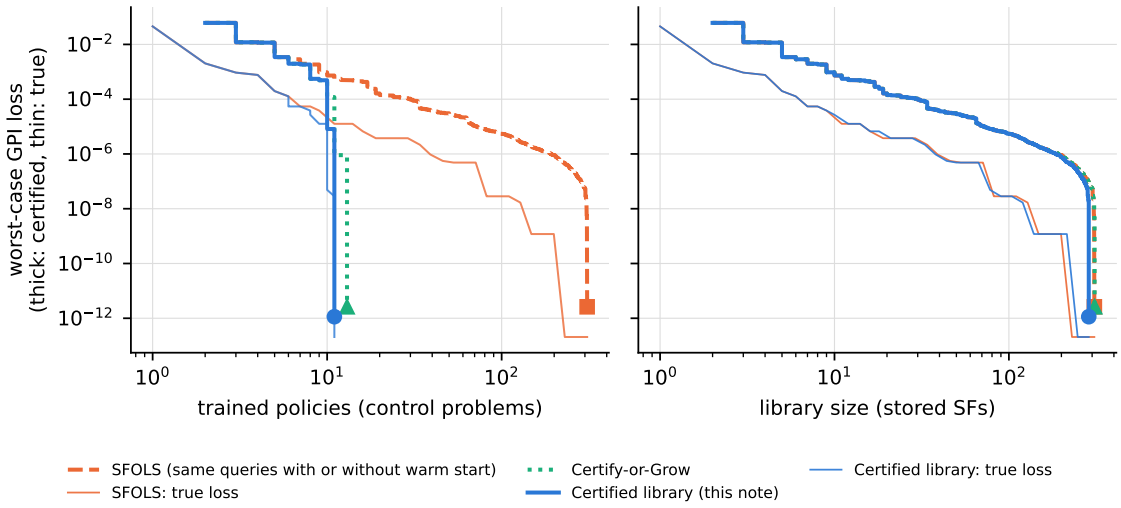


Figure 1: Double slit: anytime certificate (thick) and true worst-case loss (thin) against trained policies and library size.

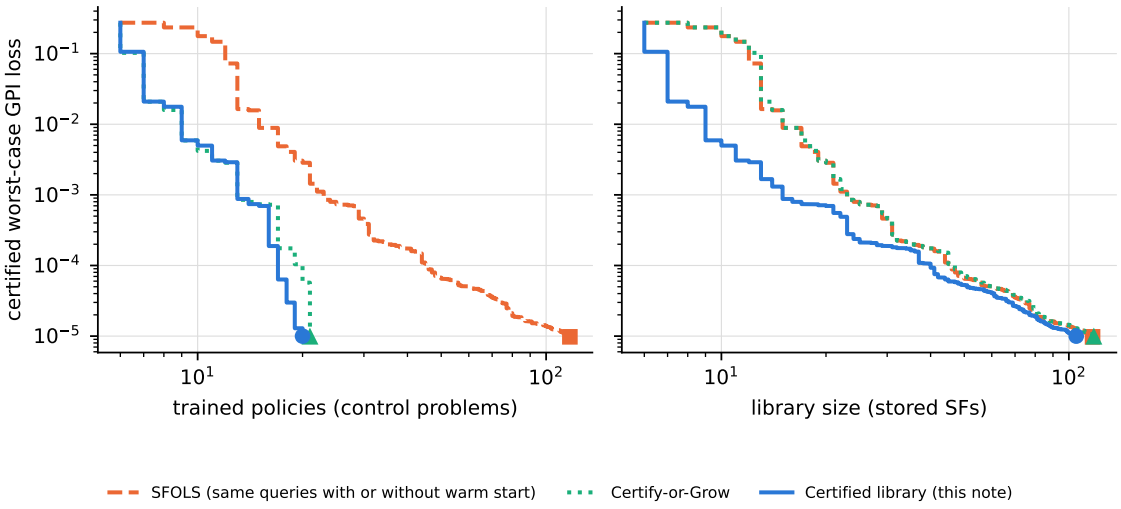


Figure 2: Windy two-stage: anytime certificate against trained policies and library size.

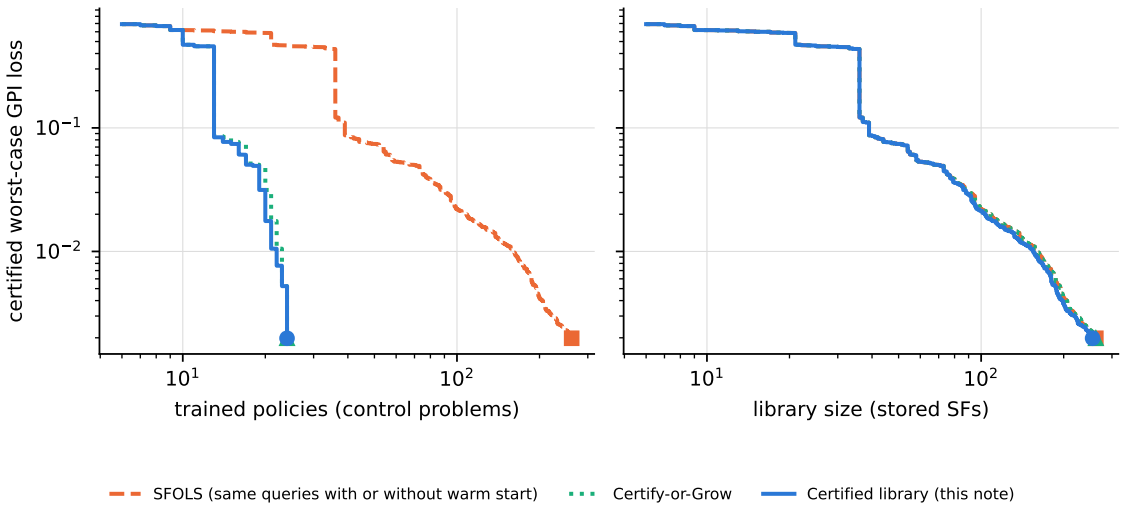


Figure 3: Office, $p = 0.1$, three 4-exit faces: anytime certificate against trained policies and library size.

Table 1: Double slit (Kuric et al., wind ± 3, $\gamma = 0.9$), weights on the whole simplex (the segment), relevant state = start. Each row is the state of one anytime run when $\Gamma \leq \epsilon$ is first certified; true loss on 2001 weights.

ϵ	method	trained	library	sweeps	certified Γ	true max loss
10^{-3}	SFOLS	9	9	3933	$9.6 \cdot 10^{-4}$	$3.8 \cdot 10^{-5}$
10^{-3}	SFOLS, warm start	9	9	1250	$9.6 \cdot 10^{-4}$	$3.8 \cdot 10^{-5}$
10^{-3}	Certify-or-Grow	8	9	1408	$9.6 \cdot 10^{-4}$	$3.8 \cdot 10^{-5}$
10^{-3}	Certified library	8	9	1496	$9.6 \cdot 10^{-4}$	$3.8 \cdot 10^{-5}$
10^{-4}	SFOLS	30	30	14134	$9.5 \cdot 10^{-5}$	$3.8 \cdot 10^{-6}$
10^{-4}	SFOLS, warm start	30	30	2185	$9.5 \cdot 10^{-5}$	$3.8 \cdot 10^{-6}$
10^{-4}	Certify-or-Grow	11	30	2396	$9.5 \cdot 10^{-5}$	$3.8 \cdot 10^{-6}$
10^{-4}	Certified library	10	30	2535	$9.5 \cdot 10^{-5}$	$3.8 \cdot 10^{-6}$
10^{-6}	SFOLS	191	191	98762	$9.8 \cdot 10^{-7}$	$1.2 \cdot 10^{-9}$
10^{-6}	SFOLS, warm start	192	192	5269	$9.8 \cdot 10^{-7}$	$1.2 \cdot 10^{-9}$
10^{-6}	Certify-or-Grow	11	192	5480	$9.8 \cdot 10^{-7}$	$1.2 \cdot 10^{-9}$
10^{-6}	Certified library	11	186	5781	$9.8 \cdot 10^{-7}$	$1.2 \cdot 10^{-9}$
exact	SFOLS	310	310	274282	$2.3 \cdot 10^{-11}$	$2.1 \cdot 10^{-13}$
exact	SFOLS, warm start	309	309	6788	$2.6 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$
exact	Certify-or-Grow	13	310	7011	$2.6 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$
exact	Certified library	11	287	7412	$1.1 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$

Table 2: Sufficiency test on double slit: the base is the first k trained policies of the certified-library run; afterwards only compositions (FREE, COMPOSE, META), no training.

trained base	open corners	library	sweeps	certified Γ	true max loss
2	42	309	7784	$2.0 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$
4	27	299	7587	$8.5 \cdot 10^{-11}$	$2.1 \cdot 10^{-13}$
6	16	294	7161	$2.0 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$
8	5	290	6958	$2.0 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$
11	0	287	7051	$2.0 \cdot 10^{-12}$	$2.1 \cdot 10^{-13}$

Table 3: Sample-based construction on double slit ($\epsilon = 10^{-4}$): n sampled outcomes per (s, a) define $\hat{P}$; the library is certified on $\hat{P}$ and deployed on the true P.

n	trained	library	Γ on $\hat{P}$	true max loss on P
10	15	28	$9.9 \cdot 10^{-5}$	$1.3 \cdot 10^{-2}$
100	14	31	$9.7 \cdot 10^{-5}$	$1.6 \cdot 10^{-3}$
1000	11	29	$9.9 \cdot 10^{-5}$	$3.7 \cdot 10^{-4}$
10000	11	31	10^{-4}	$4.5 \cdot 10^{-5}$

Table 4: Windy two-stage ($\gamma = 0.95$): the faces of every two-stage task ($\{A, B\}$ and every pair of corridor exits); relevant = start and exits; regret over 60 tasks.

ϵ	method	trained	library	sweeps	certified Γ	true max loss	max task regret
10^{-3}	SFOLS	23	23	7744	$8.5 \cdot 10^{-4}$	$1.2 \cdot 10^{-5}$	0 (0.00%)
10^{-3}	SFOLS, warm start	23	23	4032	$8.5 \cdot 10^{-4}$	$1.2 \cdot 10^{-5}$	0 (0.00%)
10^{-3}	Certify-or-Grow	13	23	4188	$8.5 \cdot 10^{-4}$	$1.2 \cdot 10^{-5}$	0 (0.00%)
10^{-3}	Certified library	13	15	2865	$8.8 \cdot 10^{-4}$	$1.3 \cdot 10^{-5}$	$6.7 \cdot 10^{-6}$ (0.00%)
10^{-4}	SFOLS	46	46	18808	$8.8 \cdot 10^{-5}$	$1.5 \cdot 10^{-7}$	0 (0.00%)
10^{-4}	SFOLS, warm start	46	46	5007	$8.8 \cdot 10^{-5}$	$1.5 \cdot 10^{-7}$	0 (0.00%)
10^{-4}	Certify-or-Grow	20	47	5442	$8.8 \cdot 10^{-5}$	$1.5 \cdot 10^{-7}$	0 (0.00%)
10^{-4}	Certified library	17	40	4214	$9.3 \cdot 10^{-5}$	$8.4 \cdot 10^{-6}$	$6.7 \cdot 10^{-6}$ (0.00%)
10^{-5}	SFOLS	117	117	53719	$9.8 \cdot 10^{-6}$	$1.1 \cdot 10^{-13}$	0 (0.00%)
10^{-5}	SFOLS, warm start	117	117	6692	$9.8 \cdot 10^{-6}$	$1.1 \cdot 10^{-13}$	0 (0.00%)
10^{-5}	Certify-or-Grow	21	118	7143	$9.8 \cdot 10^{-6}$	$1.1 \cdot 10^{-13}$	0 (0.00%)
10^{-5}	Certified library	20	105	5869	10^{-5}	$1.1 \cdot 10^{-13}$	0 (0.00%)
–	corners only	6	6	–	–	$3.4 \cdot 10^{-3}$	$3.0 \cdot 10^{-3}$ (1.19%)

Table 5: Office with slip $p = 0.1$ ($\gamma = 0.99$): FSAs whose states branch on ≤ 2 symbols (three 4-exit faces); relevant = start and exits; regret over 103 tasks.

ϵ	method	trained	library	sweeps	certified Γ	true max loss	max task regret
$2.0 \cdot 10^{-2}$	SFOLS, warm start	108	108	19980	$2.0 \cdot 10^{-2}$	$3.5 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
$2.0 \cdot 10^{-2}$	Certify-or-Grow	21	108	20618	$2.0 \cdot 10^{-2}$	$3.5 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
$2.0 \cdot 10^{-2}$	Certified library	20	103	21016	$2.0 \cdot 10^{-2}$	$3.5 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
10^{-2}	SFOLS, warm start	161	161	22746	$9.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
10^{-2}	Certify-or-Grow	23	161	23421	$9.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
10^{-2}	Certified library	22	155	23910	$9.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
$5.0 \cdot 10^{-3}$	SFOLS, warm start	195	195	24438	$4.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
$5.0 \cdot 10^{-3}$	Certify-or-Grow	24	195	25144	$4.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
$5.0 \cdot 10^{-3}$	Certified library	24	188	25769	$4.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$2.7 \cdot 10^{-6}$ (0.00%)
$2.0 \cdot 10^{-3}$	SFOLS, warm start	262	262	27848	$2.0 \cdot 10^{-3}$	$9.9 \cdot 10^{-5}$	$2.7 \cdot 10^{-6}$ (0.00%)
$2.0 \cdot 10^{-3}$	Certify-or-Grow	24	262	28554	$2.0 \cdot 10^{-3}$	$9.9 \cdot 10^{-5}$	$2.7 \cdot 10^{-6}$ (0.00%)
$2.0 \cdot 10^{-3}$	Certified library	24	255	29258	$2.0 \cdot 10^{-3}$	$9.9 \cdot 10^{-5}$	$2.7 \cdot 10^{-6}$ (0.00%)
–	corners only	6	6	–	–	$6.6 \cdot 10^{-4}$	$1.6 \cdot 10^{-5}$ (0.00%)

7 Assessment

- **What is certified.** Theorem 1 bounds, for every weight of a face union and every relevant state, both GPI's loss and the planner's error, from stored SFs and Bellman gaps only; Corollary 1 turns it into a zero-shot bound for every FSA of the class. The bound is valid but conservative (true losses are 20–800$\times$ below Γ; task regrets lower still).
- **Deterministic domains.** The d extreme policies are certified exactly optimal with no composition and a few hundred backups (Office).
- **Stochastic domains.** Training drops to the extremes plus a few policies (or only the extremes, if compositions are iterated); evaluations do not drop: the library has the size of an ϵ-CCS (Section 5), and total compute is that of warm-started SFOLS plus 5–20%, except where FREE closes corners (windy: 12–29% less).
- **Model-free.** With an empirical model the certificate holds for $\hat{P}$ only; the true loss can exceed it by the model error, so a rigorous model-free certificate needs confidence terms.
- **Scaling.** Corner enumeration is exact and cheap on edges, affordable on 4-exit faces, and expensive on the full 6-exit simplex (tens of thousands of corners per relevant state).

References

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